Joshua Rivera Camacho

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Education

Massachusetts Institute of Technology (MIT) — Cambridge, MA

B.S. in Mechanical Engineering, Minor in Design — *Expected May 2027*

- GPA: 4.4/5.0
- Relevant Coursework: Design and Manufacturing I, Thermo-Fluids Engineering I, Mechanics and Materials I, Dynamics and Controls I, Design Studio: Objects and Interaction, Interaction Intelligence, Tool Design
- Activities: MIT MAD & Metropolis Shop Mentor, DME Head & Logistics Coordinator, Assistive Tech, MIT Energy Initiative (MITEI), Phi Kappa Sigma Secretary, MIT Global Teaching Labs, MITES Semester '22

Experience

Head Coordinator — Mar. 2026 – Aug. 2026

Discover Mechanical Engineering (DME), MIT — *Cambridge, MA*

- Redesigned the program's build-challenge robot (unchanged since 2016) for 30+ incoming first-year students, many with no build experience, shortening program time by a full day to meet new scheduling constraints
- Planned and executed logistics for a three-daypre-orientation robot build challenge
- Co-managed $11K program budget, procuring supplies and meals while delivering the program under budget

Undergraduate Researcher — Feb. 2026 – Aug. 2026

Design Intelligence Lab, MIT — *Cambridge, MA*

- Designed and fabricated physically embedded AI objects for proprietary models, testing physical embodiments of AI-inferred environmental states. In collaboration with Archetype AI
- Prototyped enclosures and mechanisms using SLA & FDM printing, laser cutting, and embedded electronics, optimizing packaging to achieve target form factors

Undergraduate Researcher — Jun. 2025 – Dec. 2025

Ideation Laboratory, MIT — *Cambridge, MA*

- Co-authored research on a novel 3D printing architecture that maximizes print-bed utilization to near 100% lateral efficiency — published and awarded **Best Paper at ACM CHI 2026**
- Designed and fabricated a low-cost prototype of the novel printing architecture, enabling accessible hardware iteration without reliance on original machined assembly.

Design Chair — Apr. 2025 – Oct. 2025

Skuffle Halloween Fundraiser, Phi Kappa Sigma — *Cambridge, MA*

- Co-designed and led fabrication of Skuffle, a two-night Halloween haunted house fundraiser that raised over $7K for the Breast Cancer Research Fund

Selected Projects

Additional projects available at joshuariveracamacho.com

Pear Cat | *2.007: Design and Manufacturing I* — Feb. 2025 – May 2025

- Designed and fabricated a competition robot featuring a gear spinner on a belt-driven lift, all-wheel drive, and an actuated hauling hitch; programmed autonomous light-activated start sequence
- **Won MIT 2.007 Animatronic Oz Prize** for most complex/intricate mechanism at the class competition

FDM Beam Mechanics Study | *2.671: Measurement and Instrumentation* — Sep. 2025 – Dec. 2025

- Designed and executed three-point beam bending study on FDM-printed PLA beams to characterize the relationship of geometry, layer height, and print orientation on flexural modulus and failure load
- Presented findings as a flash talk, research paper, and poster at the class research symposium

Pool Boiling Facility | *MIT The Red Lab* — Jun. 2024 – Mar. 2025

- Designed and fabricated a pool boiling experiment allowing gas bubble injection at the point of heating, with infrared and high-speed camera recording capabilities
- Selected to present at the MIT Energy Initiative Project Symposium

Skills

Digital Design & Programming: SOLIDWORKS (CAD & FEA), Fusion (CAD & CAM), Rhino & Grasshopper, Onshape, Python, MATLAB, Arduino IDE, Excel, Adobe Illustrator
Fabrication: FDM/SLA 3D Printing, Laser Cutting, Milling, Lathing, General Machine Tools, Welding, Circuitry
Languages: English (native), Spanish (native)